

**FORMAN CHRISTIAN COLLEGE**  
**(A CHARTERED UNIVERSITY)**



**CSCS 306 - A**

**FALL 24**

**Lab 10 Report**

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## Introduction

This lab focused on creating a system to log data from an ESP32 microcontroller to a Google Sheet using Google Apps Script. The objective was to toggle an LED's state using a push-button and to record the state changes (ON/OFF) along with corresponding date and time in a cloud-based Google Sheet. The project emphasized the integration of embedded systems with cloud-based services, real-time data logging, and practical usage of ESP32 capabilities.

## Functions

### ESP32 Implementation

- **Button State Detection:**
  - Read the button state to toggle the LED state.
  - Implement debounce logic to handle false button readings.
- **Google Sheet Integration:**
  - Utilize WiFi connectivity of the ESP32 to send HTTP GET requests to a Google Apps Script Web App.
  - Include parameters: ledstatus (1 or 0), date, and time.

### Google Apps Script

- Append received parameters (ledstatus, date, time) to a Google Sheet.
- Deploy the script as a Web App with public access.

## Algorithm and Logic

### ESP32 Code

#### Setup Phase:

1. Initialize serial communication for debugging.
2. Configure GPIO pins for LED and button.
3. Connect to WiFi and synchronize time using an NTP server.

#### Loop Phase:

1. Detect button presses and toggle the LED state.
2. On state change, send an HTTP GET request with LED state, date, and time.

3. Log the response for debugging.

### Google Apps Script

1. Parse incoming parameters: ledstatus, date, and time.
2. Append the data as a new row in the Google Sheet.
3. Return a success message.

### Code Implementation

#### Arduino Code

```
#include "WiFi.h"
#include <HTTPClient.h>
#include "time.h"

const char* ntpServer = "pool.ntp.org";
const long gmtOffset_sec = 18000; // Adjust for your timezone
const int daylightOffset_sec = 3600;

const char* ssid = "Lounge";
const char* password = "redbrown7";

String GOOGLE_SCRIPT_ID = "AKfycbxR7p9LIBckfXaVeGAYhuNuU02_c4-Aao7_dL8kxvJsL-9bDxd5HwvJrmnKqpQUBVTo";

#define LED_PIN 2
#define BUTTON_PIN 15

bool ledState = false; // Current state of the LED
bool lastLedState = false; // To track state changes

void setup() {
    Serial.begin(115200);
    delay(1000);
```

```

Serial.println("\nStarting setup...");

// Configure GPIO pins
pinMode(LED_PIN, OUTPUT);
pinMode(BUTTON_PIN, INPUT_PULLUP);

// Connect to WiFi
Serial.print("Connecting to WiFi: ");
Serial.println(ssid);
WiFi.begin(ssid, password);

while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
}

Serial.println("\nWiFi connected.");
configTime(gmtOffset_sec, daylightOffset_sec, ntpServer);
Serial.println("Time synchronization initialized.");
}

void uploadToGoogleSheet(bool state) {
    if (WiFi.status() == WL_CONNECTED) {
        struct tm timeinfo;

        if (!getLocalTime(&timeinfo)) {
            Serial.println("Failed to obtain time.");
            return;
        }

        // Format date and time separately
        char dateBuff[15], timeBuff[10];
        strftime(dateBuff, sizeof(dateBuff), "%Y-%m-%d", &timeinfo);
        strftime(timeBuff, sizeof(timeBuff), "%H:%M:%S", &timeinfo);
    }
}

```

```

        // Create the URL

        String url = "https://script.google.com/macros/s/" + GOOGLE_SCRIPT_ID
+ "/exec?ledstatus=" + String(state ? "1" : "0") +
            "&date=" + String(dateBuff) +
            "&time=" + String(timeBuff);

        // Send HTTP request
        HTTPClient http;
        http.begin(url);
    }
}

void loop() {
    bool buttonState = !digitalRead(BUTTON_PIN); // Inverted logic for
INPUT_PULLUP

    // Toggle LED based on button state
    if (buttonState) {
        if (!ledState) { // Change state only if it's off
            ledState = true;
            digitalWrite(LED_PIN, HIGH);
            Serial.println("LED turned ON.");
        }
    } else {
        if (ledState) { // Change state only if it's on
            ledState = false;
            digitalWrite(LED_PIN, LOW);
            Serial.println("LED turned OFF.");
        }
    }

    // Upload only on state change

```

```

    if (ledState != lastLedState) {
        uploadToGoogleSheet(ledState);
        lastLedState = ledState;
    }

    delay(500);
}

```

## Google Apps Script

```

var sheet_id = "1g0_lcIP1TWKY2727e1yCCflBdPvNy-UK7Yg7DGv60NA";
var sheet_name = "lab10";

function doGet(e) {
    var ledStatus = e.parameter.ledstatus; // LED status: 1 (ON) or 0 (OFF)
    var date = e.parameter.date;           // Date
    var time = e.parameter.time;           // Time

    // Open the sheet
    var ss = SpreadsheetApp.openById(sheet_id);
    var sheet = ss.getSheetByName(sheet_name);

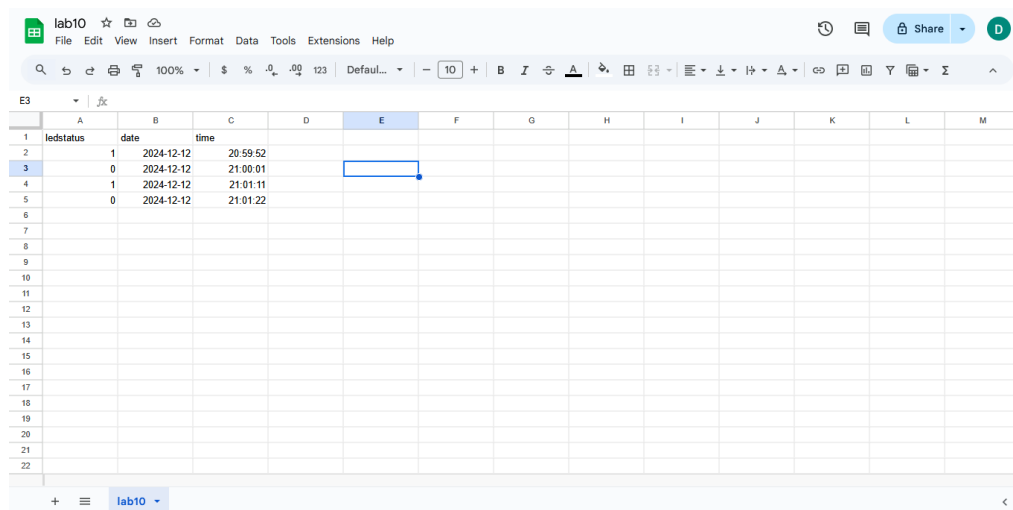
    // Append the data
    sheet.appendRow([ledStatus, date, time]);

    // Return success response
    return ContentService.createTextOutput("Success");
}

```

## Output

The system successfully logged LED state changes to the Google Sheet, as shown below:



|    | A         | B          | C        | D | E | F | G | H | I | J | K | L | M |
|----|-----------|------------|----------|---|---|---|---|---|---|---|---|---|---|
| 1  | ledstatus | date       | time     |   |   |   |   |   |   |   |   |   |   |
| 2  | 1         | 2024-12-12 | 20:59:52 |   |   |   |   |   |   |   |   |   |   |
| 3  | 0         | 2024-12-12 | 21:00:01 |   |   |   |   |   |   |   |   |   |   |
| 4  | 1         | 2024-12-12 | 21:01:11 |   |   |   |   |   |   |   |   |   |   |
| 5  | 0         | 2024-12-12 | 21:01:22 |   |   |   |   |   |   |   |   |   |   |
| 6  |           |            |          |   |   |   |   |   |   |   |   |   |   |
| 7  |           |            |          |   |   |   |   |   |   |   |   |   |   |
| 8  |           |            |          |   |   |   |   |   |   |   |   |   |   |
| 9  |           |            |          |   |   |   |   |   |   |   |   |   |   |
| 10 |           |            |          |   |   |   |   |   |   |   |   |   |   |
| 11 |           |            |          |   |   |   |   |   |   |   |   |   |   |
| 12 |           |            |          |   |   |   |   |   |   |   |   |   |   |
| 13 |           |            |          |   |   |   |   |   |   |   |   |   |   |
| 14 |           |            |          |   |   |   |   |   |   |   |   |   |   |
| 15 |           |            |          |   |   |   |   |   |   |   |   |   |   |
| 16 |           |            |          |   |   |   |   |   |   |   |   |   |   |
| 17 |           |            |          |   |   |   |   |   |   |   |   |   |   |
| 18 |           |            |          |   |   |   |   |   |   |   |   |   |   |
| 19 |           |            |          |   |   |   |   |   |   |   |   |   |   |
| 20 |           |            |          |   |   |   |   |   |   |   |   |   |   |
| 21 |           |            |          |   |   |   |   |   |   |   |   |   |   |
| 22 |           |            |          |   |   |   |   |   |   |   |   |   |   |

